C - Pointers

Basic Idea

- What is a pointer?
  - A different variable type that stores a memory address
  - Associated with a specifc varabile type, aka
    - int poitner sotres the memory addrress of an int
    - float pointers store the memory address of a float
    - char pointer sstores the memory addrress of a char

Relevant Operators

- & -> refernece, aka "adress of" operator
  - used to get the memory adress of a variable
  - using it is known as "referencing"
- * -> dereference, aka "value of " operator
  - used to get the actual value stored at the memory address pointed to
  - using itis called "dereferencing" a pointer

Declaring Pointers

```
int *ipter;
char *cptr;
float *fptr;
```

- Notice the * in front of the variable name (this is what makes a pointer)
- They don't have to have "ptr" in the name

Assigning Values

- How do I change what value is stored at the memory address pointed to by a pointer?

```
int *iptr;
int *jptr;
int a, b, c

a = 5;
b = 6;
jptr = &a;
iptr = &c;
*ipter = 100;
```

Printing Pointers

- printf command
- printing memory address:
  - need to use a different formatter: %p
- printing value stored:
  - use appropriate formatter for datatype

Be Careful

- C will let you do a lot of things
- You can easily do something other than what you intended….
- Pay attention to warnings they can help you